

mf6Voronoi Studio

Quick Start Guide

Version 3.0.1

From shapefiles to a MODFLOW 6 grid in about ten minutes



Follow these seven steps on the Example folder that ships with the application. For the full reference, see the User Manual.

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1. Before you start

mf6Voronoi Studio turns a model boundary and a handful of GIS features into an unstructured Voronoi grid for MODFLOW 6 — fine where you need detail, coarse where you do not.

This guide walks one complete pass using the Example folder installed alongside the application. Work through it once and you will have produced a real grid and a ModelMuse-ready model.

You need	Details
The application	mf6Voronoi Studio 3.0.1 (check with Help → About)
The Example folder	Shipped with the application; used throughout this guide
One projected CRS	In metres or feet. The Example data is EPSG:32631, in metres
About ten minutes	The mesh in this guide generates in roughly two seconds

The one rule that matters. Every file you import must be in the same projected coordinate system, and cell sizes are distances in that system's unit. Geographic (latitude/longitude) data is rejected.

The window

The map is on the left; a scrolling column of panels is on the right, arranged top to bottom in the order you work through them.

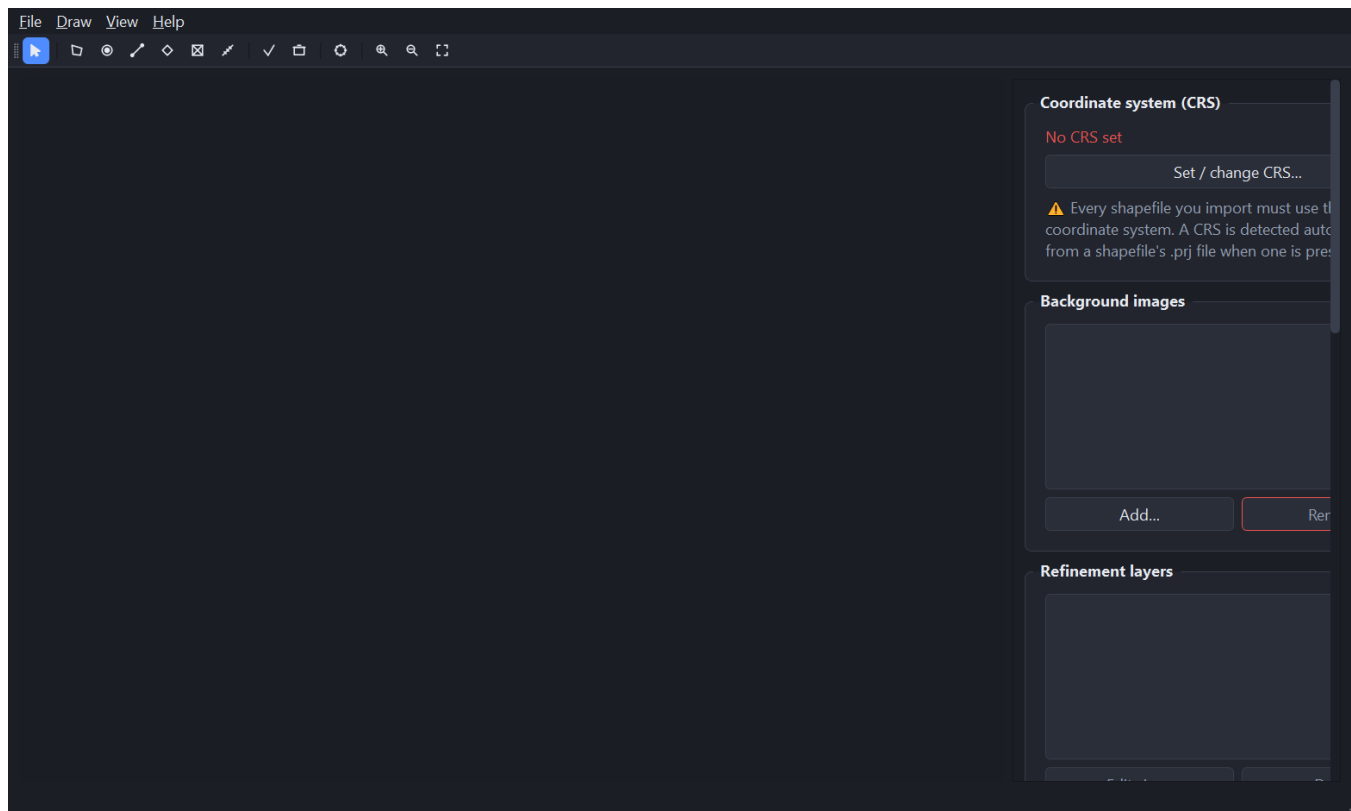


Figure 1. The application on first launch.

2. The seven steps

Step 1 — Set the coordinate system

In the **Coordinate system (CRS)** panel click **Set / change CRS...**, type **32631** and click OK. The panel should then read "WGS 84 / UTM zone 31N (EPSG:32631) — unit: meter".

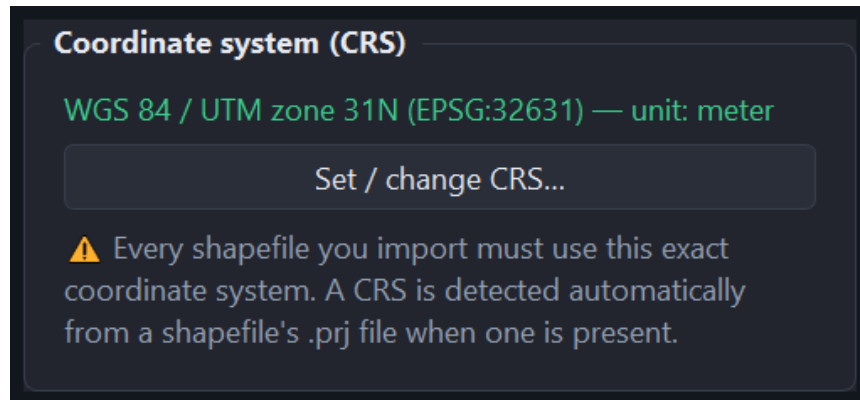


Figure 2. The CRS panel once it is set.

You can skip this step. Importing the limit polygon first picks up the CRS from its .prj file automatically.

Step 2 — Import the model limit

Choose **File → Import limit polygon...** and select **Example\modelLimit.shp**. The boundary appears in cyan and the map fits itself to it.

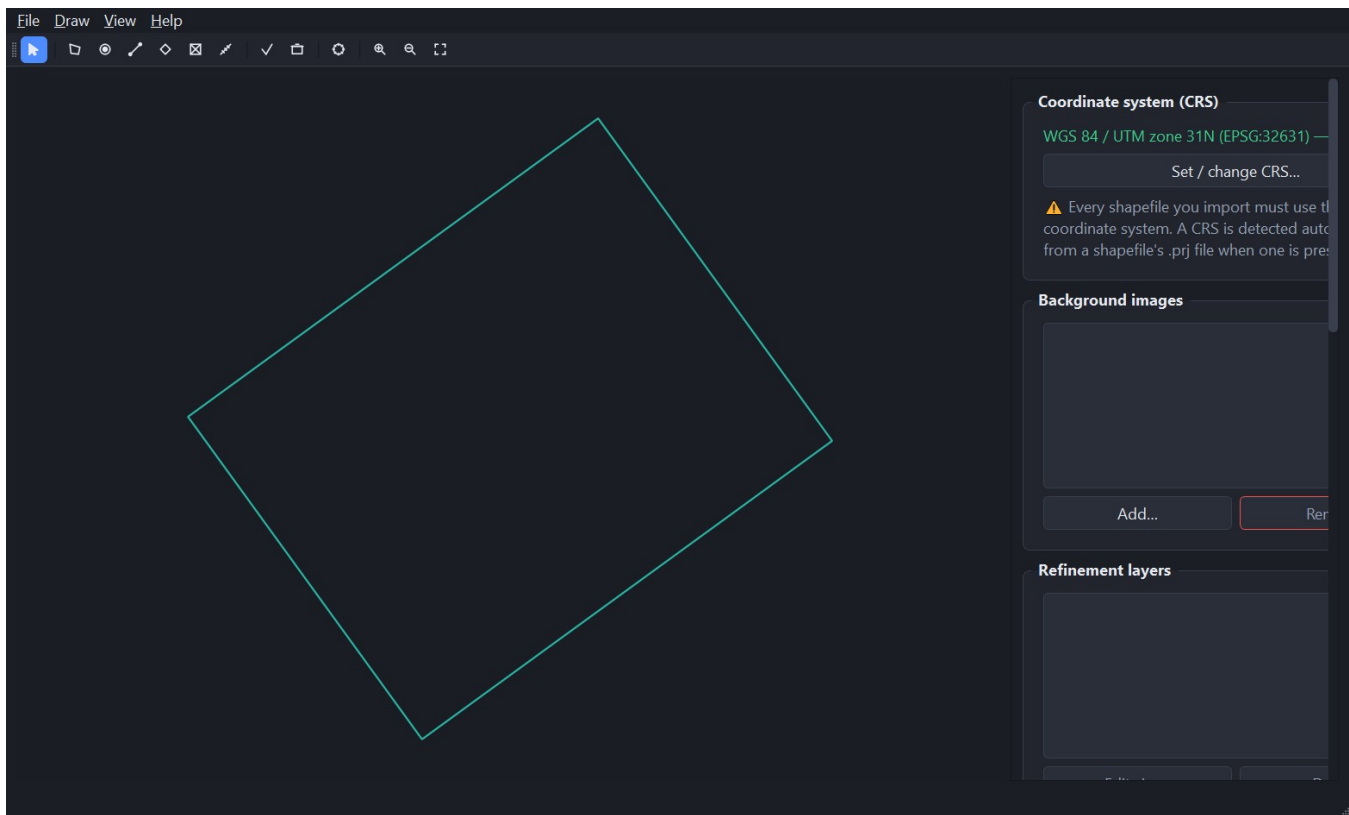


Figure 3. The model limit, loaded and fitted.

It must be **exactly one single-part polygon**. If yours is multipart, dissolve it in GIS first.

Step 3 — Add the refinement layers

These are the features the grid should follow closely. For each one, choose File → Import refinement shapefile..., pick the file, then give it a name and a target cell size.



Figure 4. Naming a layer and setting its target cell size.

Import these five, in this order — coarsest first, so cell sizes step down smoothly from the boundary to the wells:

Shapefile	Name	Target cell size
modelGhb.shp	modelGhb	10 m
site.shp	site	4 m
hfb.shp	hfb	3 m
drains.shp	drains	2 m

Shapefile	Name	Target cell size
pointWell.shp	pointWell	1 m

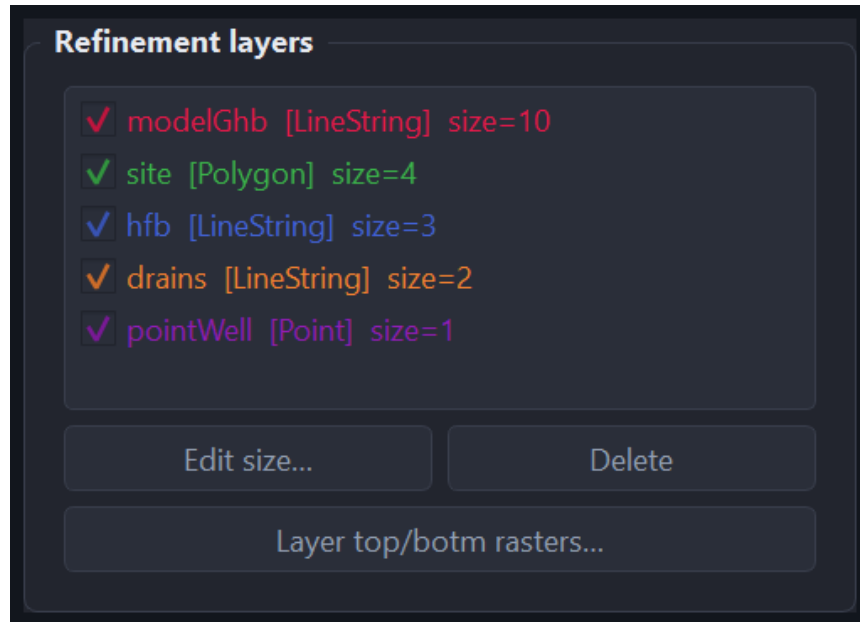


Figure 5. All five layers loaded.

You can draw layers instead of importing them — the Draw menu and the toolbar have point, line and polygon modes. Remember to click Finish layer (the check button) when you are done, or nothing is committed.

Step 4 — Set the refinement options

Scroll down to "Voronoi refinement options" and enter these four values:

Option	Value	What it controls
Coarse (max) cell size	20 m	The biggest cell, used far from every feature
Refinement multiplier	1.3	How fast cells grow outwards — about 30 % per ring
Allow overlapping refinement	On	Lets refinement zones overlap instead of competing
Fix short edges below	0.2 m	Re-seeds cells with edges shorter than this

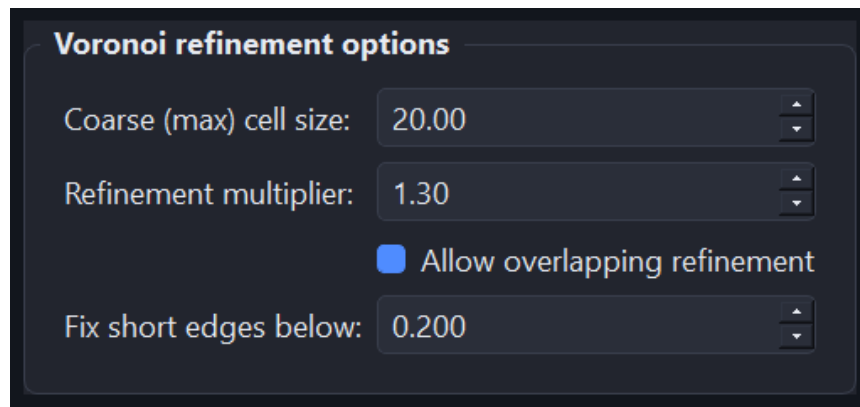


Figure 6. The options for this run.

A useful starting point on your own data: coarse size about 1/15 of the shorter side of your model, multiplier between 1.2 and 1.5.

Step 5 — Generate the mesh

Click  **Generate Voronoi mesh** in the Mesh panel (or the gear button on the toolbar), accept the default mesh name, and wait. This example finishes in about three seconds and tells you when it is done.



Figure 7. The completion popup.

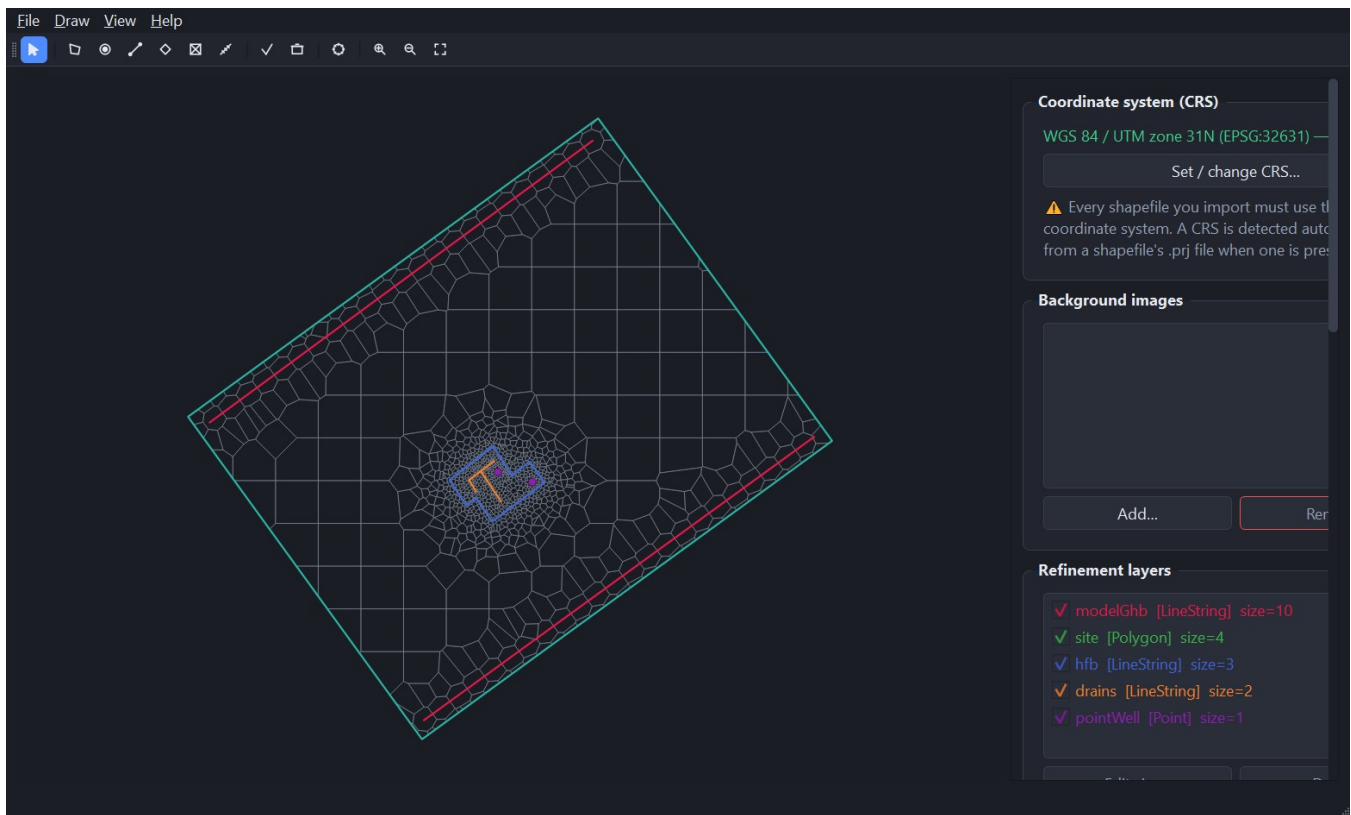


Figure 8. The result: 1,173 cells, fine at the centre, coarse towards the corners.

Step 6 — Check the mesh and the report

Zoom in on the middle of the model. The cells should tighten smoothly towards each feature — no abrupt jump from coarse to fine.



Figure 9. Refinement tightening from 20 m cells to 1 m at the two wells (magenta).

The quality report fills in on its own. Three numbers are worth a glance every time:

Check	This run	What it means
Total area	44,293.75 m ²	Must equal your model limit area. It does here, exactly — nothing was lost when the mesh was clipped to the boundary.
Cell area max	400.00 m ²	The 20 m coarse size squared, so the coarse setting was actually reached.
Cells (ncpl)	1,173	The size of your grid. If it is far larger than you expected, raise the finest target size or the multiplier.

Cell-count / quality report	
Metric	Value
Cells (ncpl)	1,173
Vertices (nvert)	0
Cell area min (meter ²)	0.16
Cell area max (meter ²)	400.00
Cell area mean (meter ²)	37.76
Cell area median (meter ²)	3.47
Total area (meter ²)	44,293.75
Vertices / cell (min)	3
Vertices / cell (max)	29
Vertices / cell (mean)	6.08
Shortest edge (meter)	0.0119
Mean of min edges (meter)	1.2231
Short-edge threshold (meter)	0.200
Cells with short edges	210
Compute report Save report...	

Figure 10. The full report for this run.

Do not panic at "Cells with short edges". The count is relative to the threshold you set. On this same mesh a 0.5 m threshold flags 56 % of cells; the 0.2 m used here flags 18 %. Set it to a small fraction of your finest target cell size and watch it as a trend between runs, not as a pass or fail.

Step 7 — Export and save

To get	Use	Result
The grid in GIS	File → Export mesh shapefile...	An ESRI polygon shapefile with a .prj
The raw DISV arrays	File → Export DISV properties (JSON)...	ncpl, nvert, vertices, cell2d, centroids
A MODFLOW 6 model	File → Export ModelMuse MODFLOW 6 model...	Eight files including mfsim.nam and the .disv
Your work back tomorrow	File → Save project (Ctrl+S)	One .mf6vor file holding everything, mesh included

The ModelMuse export in brief

1. Choose File → Export ModelMuse MODFLOW 6 model...
2. Set the number of layers to 4.
3. Set the model top to `modelTop.tif` and the four layer bottoms to `bottomLy1.tif–bottomLy4.tif` (click Raster... on each row).
4. Pick an empty output folder and click OK.
5. In ModelMuse: **File** → **Import** → **Import MODFLOW-6 Model...** → select `mfsim.nam`.

Number of layers: 4

MODFLOW 6 executable: mf6

Model top (constant or raster)

e_Code\mf6voronoi\Example\modelTop.tif aster. ×

Layer bottoms (each: constant value or raster)

Layer	Bottom elevation
-50	voronoi\Example\bottomLy1.tif aster. ×
Layer 2	voronoi\Example\bottomLy2.tif aster. ×
Layer 3	voronoi\Example\bottomLy3.tif aster. ×
Layer 4	voronoi\Example\bottomLy4.tif aster. ×

Tip: type a number for a flat surface, or click **Raster...** to sample a GeoTIFF at each cell centroid.
 In ModelMuse: **File** → **Import** → **Import MODFLOW-6 Model** → select the generated `mfsim.nam`.

OK Cancel

Figure 11. The export dialog, set up for four layers with a raster for each surface.

Any surface can be a plain number instead of a raster — type it in the field rather than browsing for a file.

3. At a glance

Toolbar

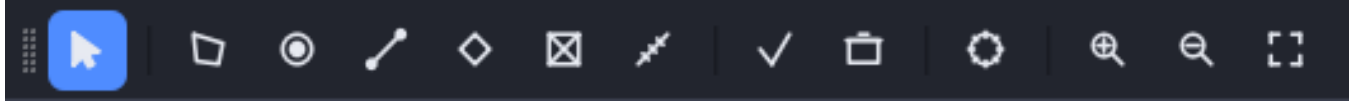


Figure 12. Left to right: Navigate · Limit · +Points · +Lines · +Polygons · +Void · Measure · Finish layer · Cancel layer · Generate mesh · Zoom in · Zoom out · Fit to data.

The first seven are modes: one is active at a time, and its button stays pressed. Navigate is the default.

Keyboard and mouse

Input	Does
Ctrl+N / Ctrl+O / Ctrl+S	New project / Open project / Save project
Ctrl+Shift+S	Save project as...
Enter	Finish the shape you are drawing
Esc	Cancel the shape or measurement in progress, otherwise deselect
Delete	Remove the selected background image or void polygon
Left-drag	Pan the map (in Navigate mode)
Middle-drag	Pan the map (in any mode)
Mouse wheel	Zoom in and out
Double-click	Finish the shape you are drawing

Five things that catch people out

- **Drawn features vanish.** They are not committed until you click Finish layer (the check button).
- **A shapefile is refused.** Its CRS differs from the project's. Re-project it in GIS — the application will not do it for you.
- **The limit will not load.** It must be exactly one single-part polygon.
- **Vertices (nvert) shows 0.** Expected — it stays 0 until you export the DISV properties or a ModelMuse model.
- **Far more cells than expected.** Your finest target size is very small relative to the coarse size. Halving a target size roughly quadruples the cells in that zone.

Where to go next

The User Manual covers every menu command, every sidebar panel, the drawing tools, void polygons, background images — including the new "Online satellite map" tab for fetching a georeferenced satellite basemap directly from the internet — the measure tool, how to read each quality metric, and a troubleshooting table.

mf6Voronoi Studio is a graphical front-end for the mf6Voronoi package by Hatari Labs:
<https://github.com/hatarilabs/mf6Voronoi>

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